

HW 03

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Exercise 1

Tools and resources used:

Exercise 1 **Points attempted: 2.** **Time spent: 5 minutes.** **Difficulty: 1/10.** **Self-assessed score: 1.** **Questions/Feedback: None.**

Choice Exercise 2 **Points attempted: 2.** **Time spent: 5 minutes.** **Difficulty: 1/10.** **Self-assessed score: 2.** **Questions/Feedback: None.**

Choice Exercise 3 **Points attempted: 6.** **Time spent: 30 minutes.** **Difficulty: 3/10.** **Self-assessed score: 6.** **Questions/Feedback: Mathematical correctness, clarity.**

Exercise 2

Done! I discussed my solution to HW02 Exercise 9.4.

Exercise 3

1. ‘ ‘ Let $(G, *) = (\mathbb{Z}, +)$, and let $a = 1 \in G$ and $b = 3 \in G$. Compute $a^4 * b^{-2} \in \mathbb{Z}$.

We see that $a^4 * b^{-2}$ expands to $1 + 1 + 1 + 1 - 3 - 3 = -2$.

2. Let $g \in G$, and let $\langle g \rangle$ be the cyclic subgroup of G generated by g . Prove that the order of the element g is equal to the order of the subgroup $\langle g \rangle$.

Proof. First, we notice that the order of $\langle g \rangle$ is countably infinite. Thus, traditional counting techniques may not be used.

We can rewrite all elements of $\langle g \rangle$ as $\{g^n \mid n \in \mathbb{Z}\}$. We can also rewrite the subgroup generated by g as $\{g^n \mid n \in \mathbb{Z}\}$.

Therefore, these two group are exactly the same, and have exactly the same order. $\square$

3. Consider the cyclic group C_{20} , where r is an 18° rotation, and thus $|r| = 20$. Find all elements $r^n \in C_{20}$ such that $\langle r^n \rangle = C_{20}$.

For all $n \in \{1, 3, 7, 9, 11, 13, 17, 19\}$, $\langle r^n \rangle = C_{20}$.

4. Give an example of an infinite group $(G, *)$ and a subset $H \subseteq G$ which is closed under $*$, but where H is not a subgroup of G .

Let $H = 2\mathbb{Z} + 1$. H is closed under $*$ but does not include an identity element, so it is not a subgroup of G .

5. Prove that if $a \equiv a' \pmod{n}$, $b \equiv b' \pmod{n}$, and $a + b \equiv c \pmod{n}$, then

$$a' + b' \equiv c \pmod{n}.$$

Proof. By definition, $a - a' = kn$ and $b - b' = ln$. If $a + b \equiv c \pmod{n}$, then $a + b - c = jn$.

With some algebra, we find

$$a = kn + a', b = ln + b'$$

.

Then, we write $kn + a' + ln + b' - c = jn$. We can then subtract over some terms, then factor to get

$$a' + b' - c = jn - kn - ln$$

$$a' + b' - c = (j - k - l)n$$

.

Therefore, $a' + b' \equiv c \pmod{n}$. $\square$

Choice Exercise 6 [12]

- [4] Prove that the set of 3×3 matrices that have one ± 1 per row and column forms a group under matrix multiplication. This group is called the octahedral group, sometimes written B_3 . Without looking it up, determine the order of this group.

Proof. To prove that a given set of matrices forms a group under matrix multiplication, we must prove that it satisfies the group axioms.

First, associativity is inherited from matrix multiplication. Then, we see that the set of 3×3 matrices that have one ± 1 per row and column includes the identity matrix I .

Also, we see that this set of matrices must have linearly independent columns, because each of the columns has an element in a unique row. Thus, the matrices are invertible.

The most difficult part of this proof is proving that this set of matrices is closed under matrix multiplication. First, let $a, b \in B_3$. Let the 3 rows of a be the vectors a^1, a^2, a^3 (this non-standard notation, however it is helpful for this proof), and the 3 columns of b be the vectors b_1, b_2, b_3 .

Notice that having “one ± 1 per row and column” implies that the the vectors a^1, a^2, a^3 each must have its non-zero element be in a unique index. That is, if $a^1 = (1, 0, 0)$, then a^2, a^3 can’t be $(1, 0, 0)$ or $(-1, 0, 0)$. This same property applies to b_1, b_2, b_3 .

Now, let $c = a \times b$. I will denote each element of c as c_j^i , for $i, j \in \{1, 2, 3\}$, where i is the row index and j is the column index. By definition of matrix multiplication, $c_j^i = a^i \cdot b_j$ (dot product). The only time this value is non-zero is if the the index of the non-zero element in a^i and b_j match.

Because the index of the non-zero element of the rows of a and the columns of b are unique, if $c_j^i = a^i \cdot b_j \neq 0$, all other elements of c in the same row/col of c_j^i must be 0.

Therefore, the set of 3×3 matrices that have one ± 1 per row and column is closed under matrix multiplication, so, they form a group B_3 . $\square$

- Done!

```
module blob() {
  cylinder(20,10,5);
  translate([-5,0,10]) rotate([45,0,0]) cube([20,10,5]);
}

blob();
```

- [4] Apply all possible matrices from part 1 to this using multmatrix(...). (You probably want to write a program that prints OpenSCAD code instead of doing this by hand.)

```
def mats():
  for i in range(0, 3):
    for j in range(0, 3):
      for k in range(0, 3):
        if i == j or j == k or k == i:
          continue
        for mod in range(2 ** 3):
          mat = [[0]*3 for _ in range(3)]
          mat[0][i] = 1 if mod & 1 == 0 else -1
          mat[1][j] = 1 if mod & 2 == 0 else -1
          mat[2][k] = 1 if mod & 4 == 0 else -1

          yield mat

def det_3x3(mat):
```

```

    a, b, c = mat[0]
    d, e, f = mat[1]
    g, h, i = mat[2]
    return a*(e*i - f*h) - b*(d*i - f*g) + c*(d*h - e*g)

def color(det):
    if det == 1:
        return 'color("red")'
    return 'color("green")'

pre = """
module blob() {
  cylinder(20,10,5);
  translate([-5,0,10]) rotate([45,0,0]) cube([20,10,5]);
}

blob();
"""

head = """
multimatrix(
"""

tail = """
) { blob(); }
"""

print(pre)
for mat in mats():
    print(color(det_3x3(mat)) + head + str(mat) + tail)

```

4. [1] Do this again, but color the parts whose corresponding matrix has determinant 1 different from the color of those whose corresponding matrix has determinant -1 .

See above.

5. [1] Post on the #hw-prompt channel on Discord with a screenshot of blob, a screenshot of all of the rotations of blob, and an STL export of both.

Done!.

Choice Exercise 8 [4]

1. Write down the (trivial) subgroup of order 1 and the (improper) subgroup of order 8, and say what these mean geometrically.

The trivial subgroup would be $\{e\}$. The improper subgroup is just

$$D_4 = \{e, r, r^2, r^3, f, rf, r^2f, r^3f\}.$$

The trivial subgroup is just the symmetry of ‘not moving’ it. Then, D_4 is all the rotations and flips of a square such that it looks the same.

2. The five subgroups of order 2 would be:

- (a) $\{e, r^2\}$: rotating by 180 degrees.
- (b) $\{e, f\}$: flipping left to right.
- (c) $\{e, rf\}$: rotate CW then flip.
- (d) $\{e, r^2f\}$: rotate 180 then flip.
- (e) $\{e, r^3f\}$: rotate CCW then flip.

3. Write down the three subgroups of order 4, and describe in words what they mean geometrically.

- (a) $\{e, r, r^2, r^3\}$: Rotational symmetries.
- (b) $\{e, r^2, f, r^2f\}$: Combinations of 180 rotations and flips.
- (c) $\{e, r^2, rf, r^3f\}$: Combinations of 180 rotations, and rotation ± 90 degrees followed by a flip.

4. What do you notice about the orders of the subgroups?

The orders are factors of the order of D_4

5. Are any of these subgroups “essentially similar” to each other? If so, which ones?

I would not say that any of these are “essentially similar” because they all cover different elements of D_4 .